

Advanced (Habanero)

- Q1.** What is the slope of the line $y = 2x + 3$?
(A) 1
(B) 2
(C) 3
(D) 4
(E) 5
- Q2.** A line passes through points (2,3) and (4,5). What is the slope?
(A) 0.5
(B) 1
(C) 1.5
(D) 2
(E) 2.5
- Q3.** What is the slope of a line with equation $x = 4$?
(A) 0
(B) 1
(C) undefined
(D) -1
(E) 2
- Q4.** The slope of a line is -2 . What is the rate of change?
(A) -2 units/output
(B) 2 units/output
(C) -1 unit/output
(D) 1 unit/output
(E) 0 units/output
- Q5.** A line has a slope of 0. What is the rate of change?
(A) 0 units/output
(B) 1 unit/output
(C) -1 unit/output
(D) 2 units/output
(E) -2 units/output
- Q6.** The graph of a line shows a slope of 3. What is the rate of change?
(A) 3 units/output
(B) -3 units/output
(C) 1 unit/output
(D) -1 unit/output
(E) 0 units/output
- Q7.** What is the slope of a line that passes through (1,2) and (3,4)?
(A) 1
(B) 2
(C) $1/2$
(D) $3/2$
(E) 1
- Q8.** The slope of a line is $1/2$. What is the rate of change?
(A) $1/2$ unit/output
(B) $-1/2$ unit/output
(C) 2 units/output
(D) -2 units/output
(E) 0 units/output

Open Ended Questions (FRQ)

- Q9.** Find the slope of the line $y = -2x + 1$ and explain its meaning in the context of rate of change.
- Q10.** A water tank is being filled at a rate of $2 \cdot m^3/h$. If the tank is currently $4 \cdot m^3$ full, what is the slope of the graph that represents the volume of water in the tank over time? Explain your answer.

Chef's Tips

- Provide clear instructions and examples before the test.
- Allow students to ask questions before starting the test.
- Make sure students have the necessary materials and tools.